Case Study: 22% Energy Reduction in a 1.2 Mn Sq Ft Campus

A Bengaluru technology campus and the 14 measures that delivered it in 18 months

Background. A 1.2 million sq ft campus with six buildings, 9,500 seats and a 1,400 TR chilled water plant was consuming 31.6 million kWh a year. Energy was the second largest operating cost after manpower. The facilities team set a target of 20% reduction with a payback under three years and no impact on comfort scores.

Approach

1. Metering first: 112 sub-meters were added over six weeks, giving a 15-minute view of chillers, AHUs, lighting and the data room.
2. Baselining: three months of data normalised for weather and occupancy produced a clear picture of idle load.
3. Prioritising: measures were ranked by payback and risk; operational changes came before capital works.

Measures and results

Measure	Capex (Rs lakh)	Annual saving (kWh)	Payback
Chiller sequencing and condenser water reset	4	1,480,000	3 months
Chilled water delta-T recovery (coil cleaning, valve fixes)	9	1,120,000	7 months
AHU schedules aligned to real occupancy	2	860,000	2 months
VFDs on secondary pumps and cooling tower fans	38	1,310,000	19 months
LED retrofit with occupancy sensors, common areas	72	980,000	22 months
Data room containment and setpoint raise to 24 deg C	26	640,000	17 months
Solar rooftop, 600 kWp (PPA, no capex)	0	780,000	immediate

Outcome

Total energy fell from 31.6 to 24.6 million kWh (22%), avoided cost about Rs 5.6 crore a year at an average tariff of Rs 8 per kWh, and carbon avoided about 5,700 tCO2e. Comfort complaints fell 12% because the delta-T recovery fixed long-standing hot spots. Capex totalled Rs 1.51 crore, a blended payback of 11 months.

Lessons

- Operational fixes delivered 58% of the savings for 10% of the spend
- Trend logs convinced the finance team faster than any estimate
- Tenant communication on schedule changes mattered as much as the engineering